SCIENCEUTSAV®
Science is awesome

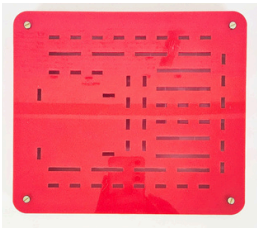
I'm a ROBO SCIENTIST

Motorized Motion and Mechanical Illusions

Session

10

STRUCTURE AND FASTENERS OF MICROCONTROLLERS

1


2


3


4


5

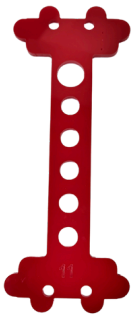

6


7


8


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11


12


straw

1


Spacers

2


L Clamp

3


Spacer Screws

4


Nuts

5


Screw

6

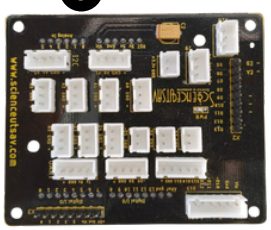

Washer

7


zinc wire 1M

8


screw hook

9


Arduino With Shield

10


Arduino cable

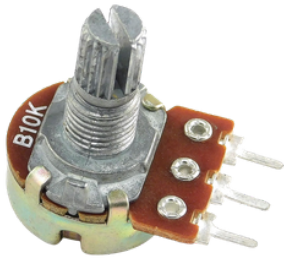
ELECTRONIC COMPONENTS



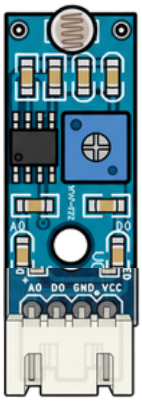
Touch sensor
A touch sensor is an electronic device that detects human touch and works like a switch. It senses the change in capacitance when a finger touches it and sends a signal to the microcontroller. This signal can be used to perform actions like turning on an LED or activating a device. Touch sensors are commonly used in mobile screens, touch lamps, and smart systems.



DC motor
A DC motor is an electrical device that converts electrical energy into mechanical energy.
It rotates when current flows through it and is used in fans, toys, and robotics projects.



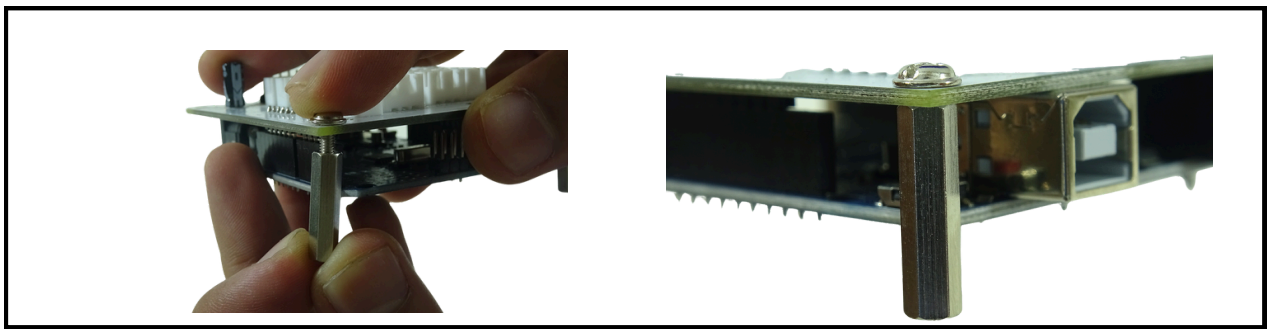
Potentiometer (POT)
A potentiometer (POT) is an electronic component used to vary resistance in a circuit.
It has three terminals and works by rotating its knob to change the voltage or current.
It is commonly used to control volume, speed, and brightness in electronic devices.



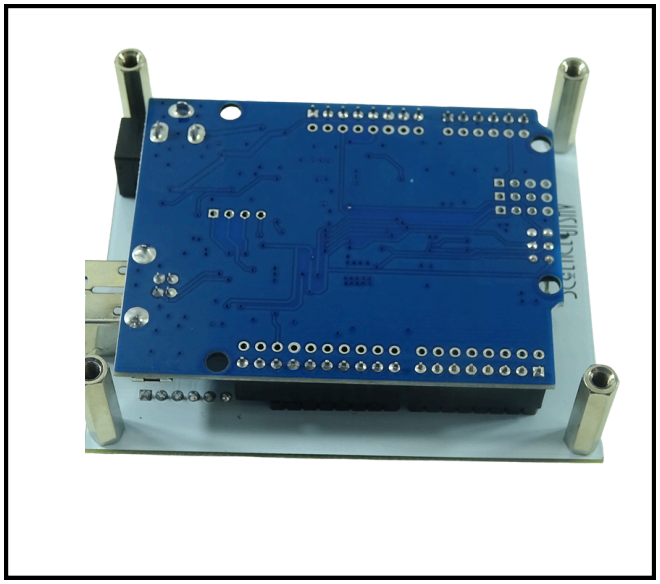
LDR sensor
An LDR sensor (Light Dependent Resistor) is a sensor that detects the intensity of light.
Its resistance changes based on the amount of light falling on it.
It is commonly used in automatic street lights, darkness detectors, and light-sensing projects.

Procedure :

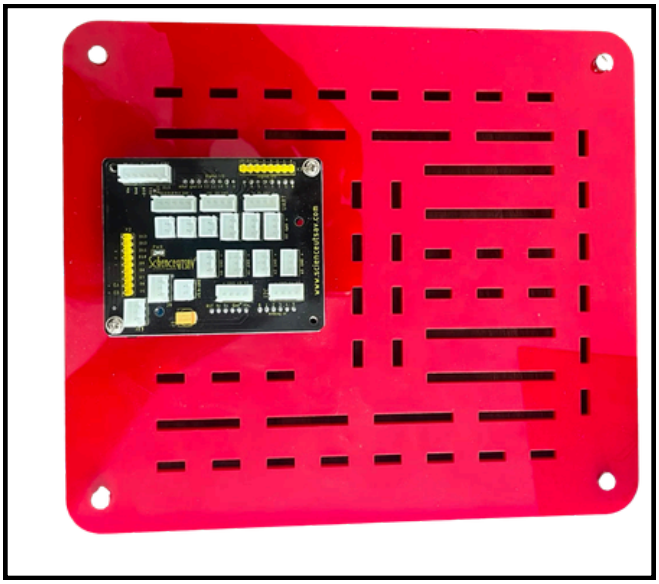
Step 1 : Fit the spacer to Arduino Shield as shown below using Spacer screw



Step 2 : Repeat the step 1 for all 4 corners of Arduino Shield



Step 3 : Place the Arduino Shield to the base board as shown below

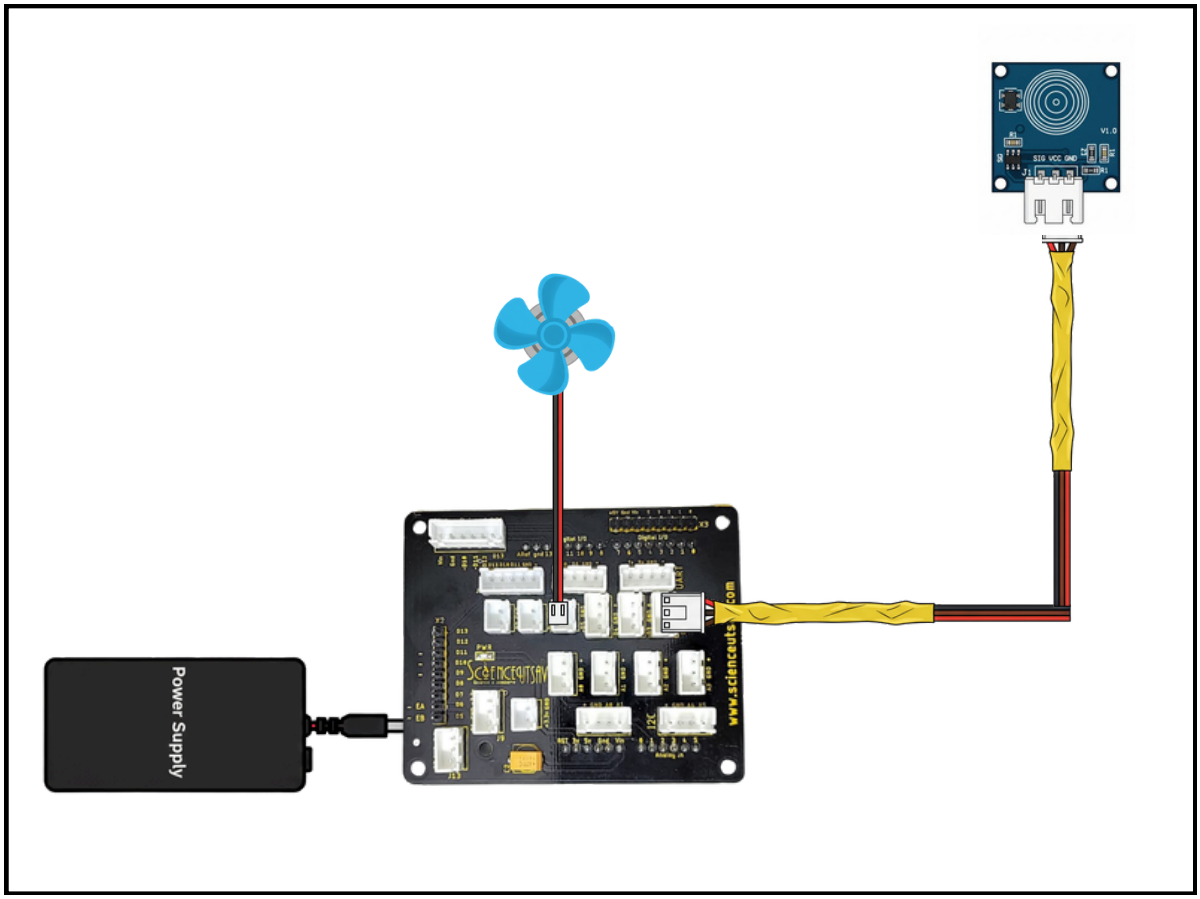


Step 4 : Fix it with the spacer screw from the bottom as shown below



BEGINNER PROJECT

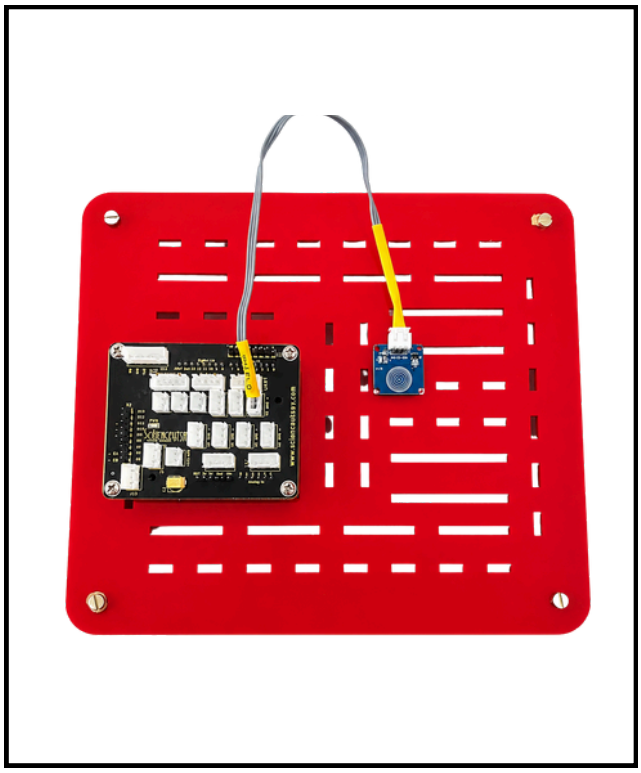
The circuit diagram of the project is given below



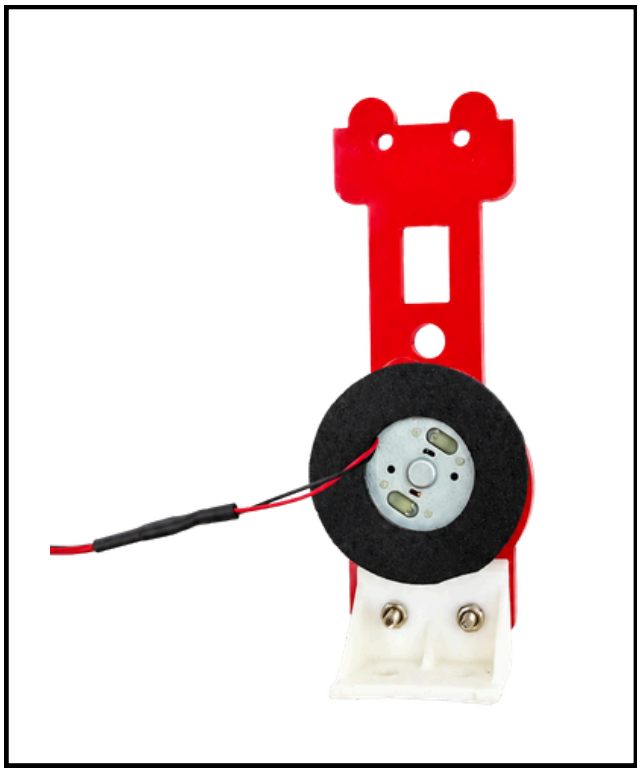
CONNECTIONS:

1. DC motor-D5
2. Touch sensor-D2

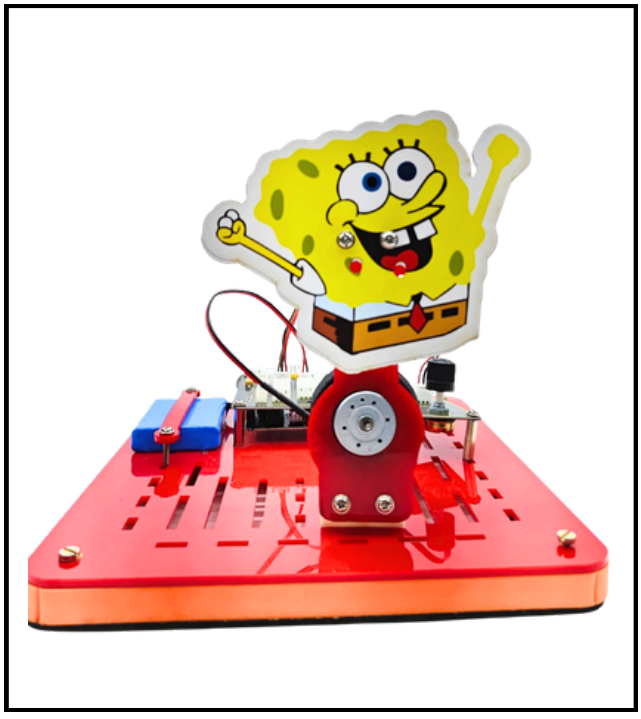
Step 1- Connect the touch sensor to the board



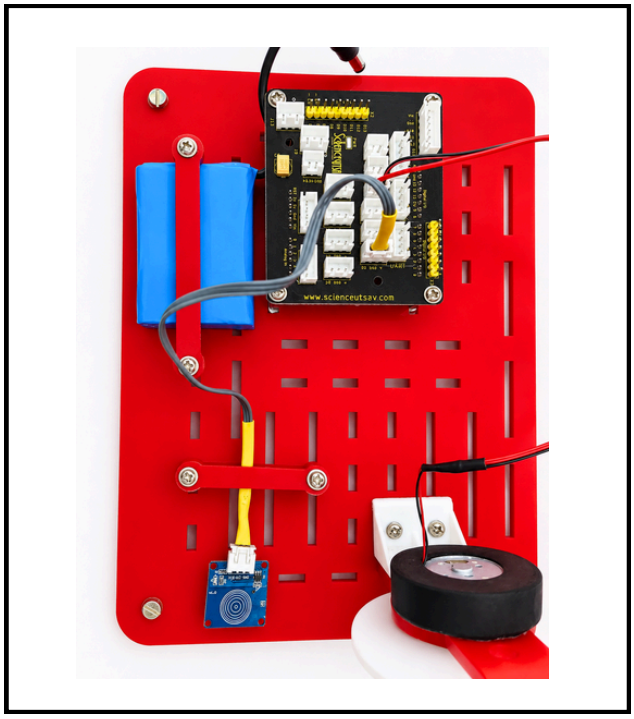
Step 2- Attach the DC motor to the stand



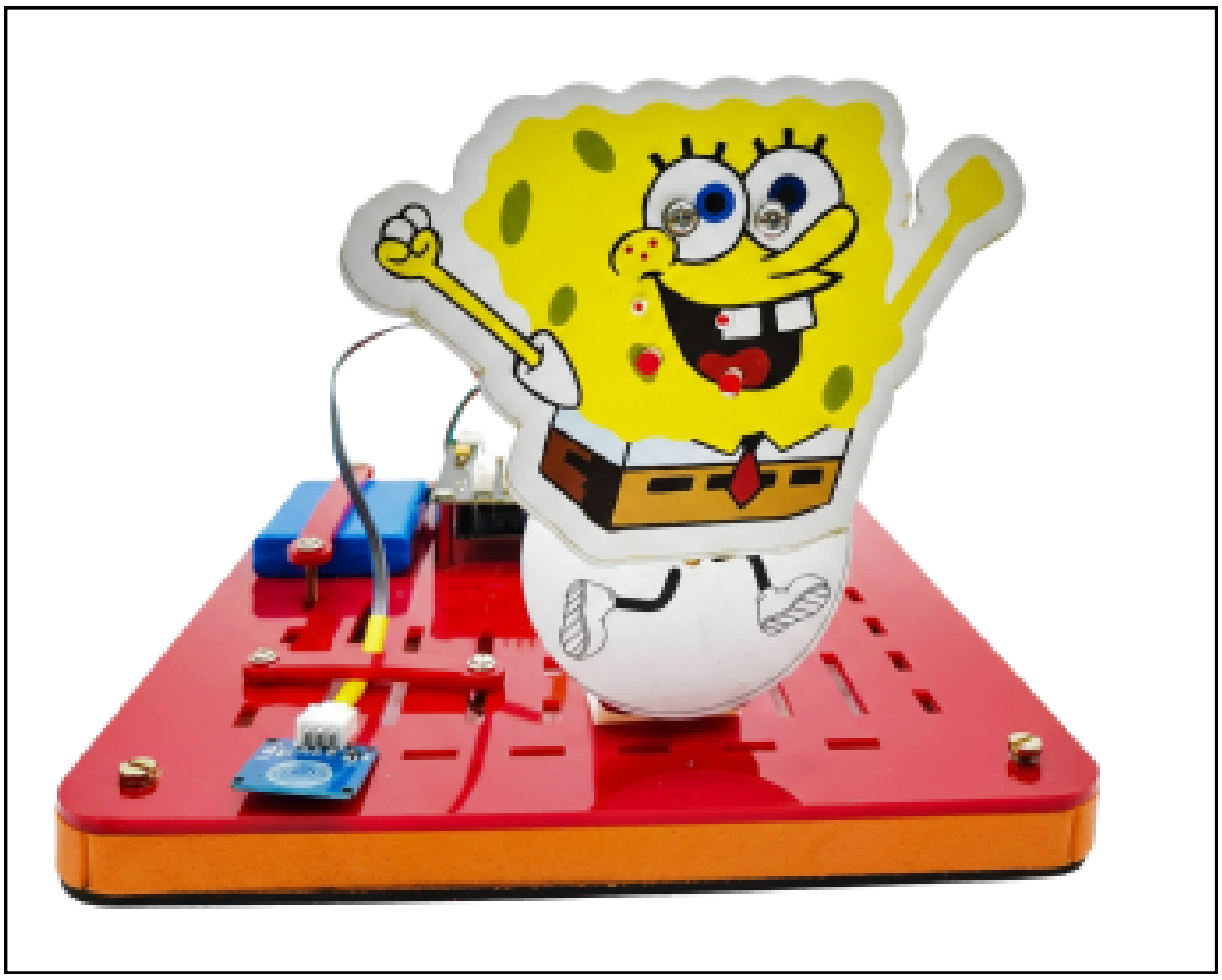
Step 3- Attach the template to the stand as shown



Step 4- Secure the touch sensor to the board using bolts



Step 5- Upload the code and observe



Code

```
/*
ScienceUtsav Arduino Code
Touch Controlled Motor Speed

Touch Sensor = D2
DC Motor PWM Pin = D5
*/

const int touchSensor = 2;
const int dcMotor = 5;

int speedLevel = 0;
bool lastTouchState = LOW;

void setup() {
  pinMode(touchSensor, INPUT);
  pinMode(dcMotor, OUTPUT);

  Serial.begin(9600);
}

void loop() {
  bool touchState = digitalRead(touchSensor);

// Detect one touch at a time
 if (touchState == HIGH && lastTouchState == LOW) {
  speedLevel++;

  if (speedLevel > 3) {
   speedLevel = 0;
  }
```

```
    delay(300);
  }

  lastTouchState = touchState;

  if (speedLevel == 0) {
    analogWrite(dcMotor, 0);      // Motor OFF
    Serial.println("Motor OFF");
  }

  if (speedLevel == 1) {
    analogWrite(dcMotor, 27);     // Slow speed
    Serial.println("Slow Speed");
  }

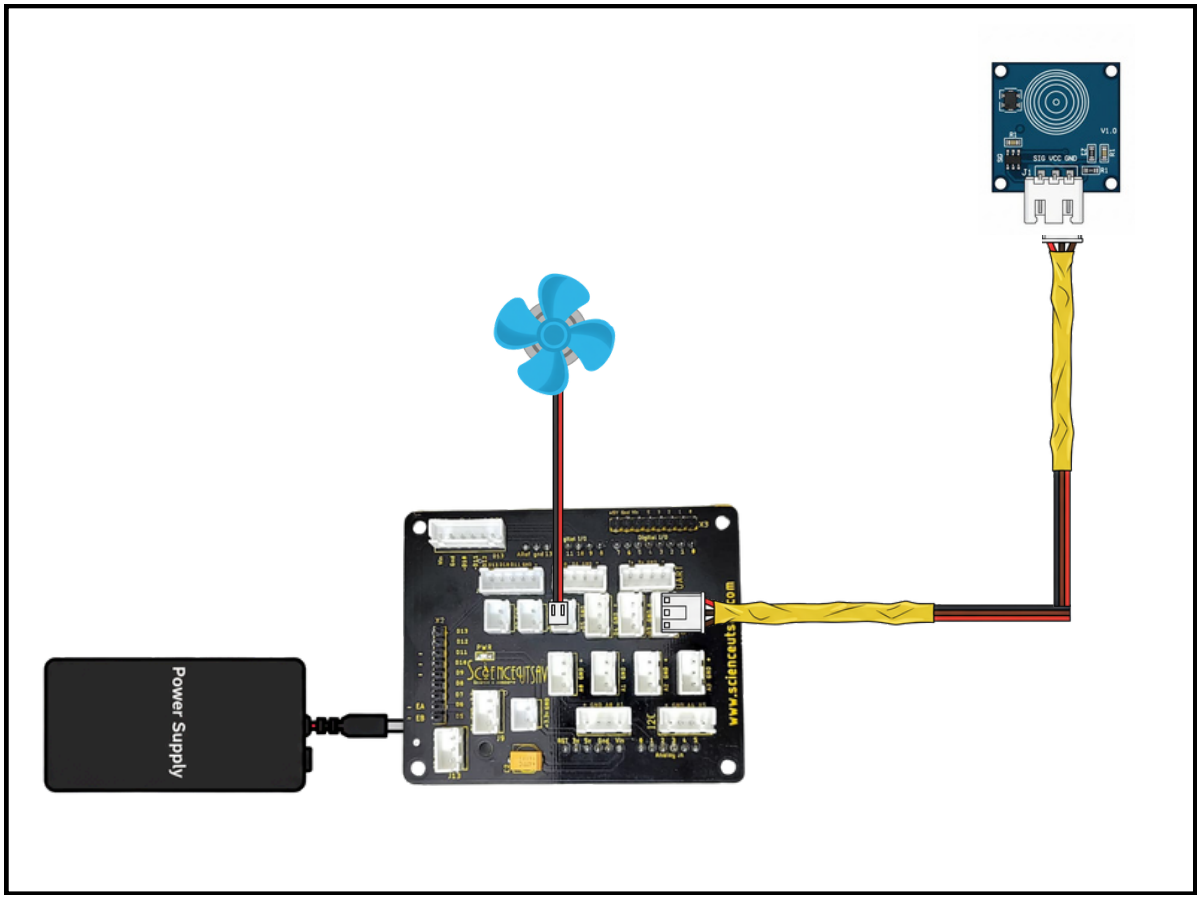
  if (speedLevel == 2) {
    analogWrite(dcMotor, 30);    // Medium speed
    Serial.println("Medium Speed");
  }

  if (speedLevel == 3) {
    analogWrite(dcMotor, 33);    // High speed
    Serial.println("High Speed");
  }

}
```

BEGINNER PROJECT

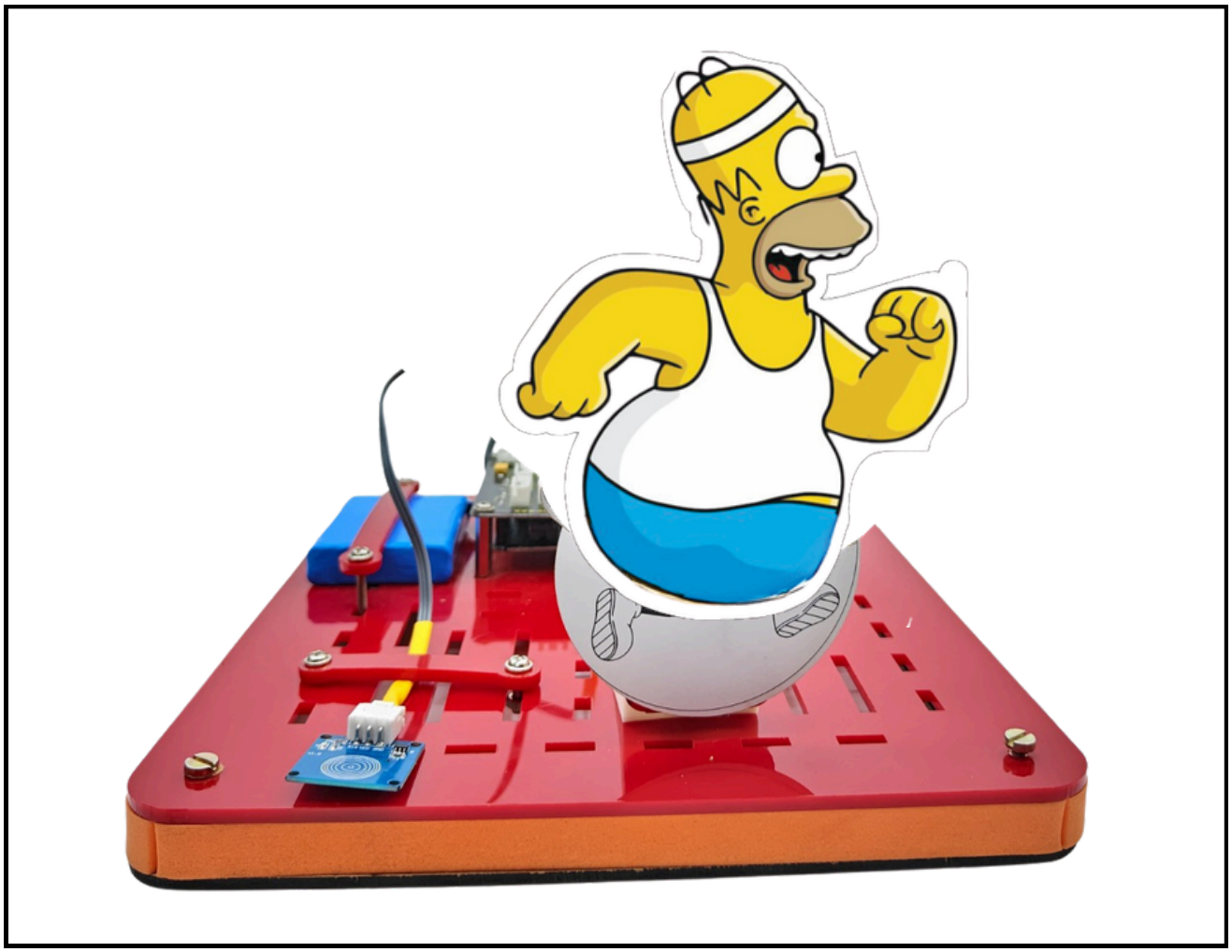
The circuit diagram of the project is given below



CONNECTIONS:

1. DC motor-D5
2. Touch sensor-D2

Follow the same procedures as above and replace the template



Code

```
/*
ScienceUtsav Arduino Code
Touch Controlled Motor Speed

Touch Sensor = D2
DC Motor PWM Pin = D5
*/

const int touchSensor = 2;
const int dcMotor = 5;

int speedLevel = 0;
bool lastTouchState = LOW;

void setup() {
  pinMode(touchSensor, INPUT);
  pinMode(dcMotor, OUTPUT);

  Serial.begin(9600);
}

void loop() {
  bool touchState = digitalRead(touchSensor);

// Detect one touch at a time
 if (touchState == HIGH && lastTouchState == LOW) {
  speedLevel++;

  if (speedLevel > 3) {
   speedLevel = 0;
  }
```

```
 delay(300);
 }

  lastTouchState = touchState;

  if (speedLevel == 0) {
    analogWrite(dcMotor, 0);      // Motor OFF
    Serial.println("Motor OFF");
  }

  if (speedLevel == 1) {
    analogWrite(dcMotor, 27);     // Slow speed
    Serial.println("Slow Speed");
  }

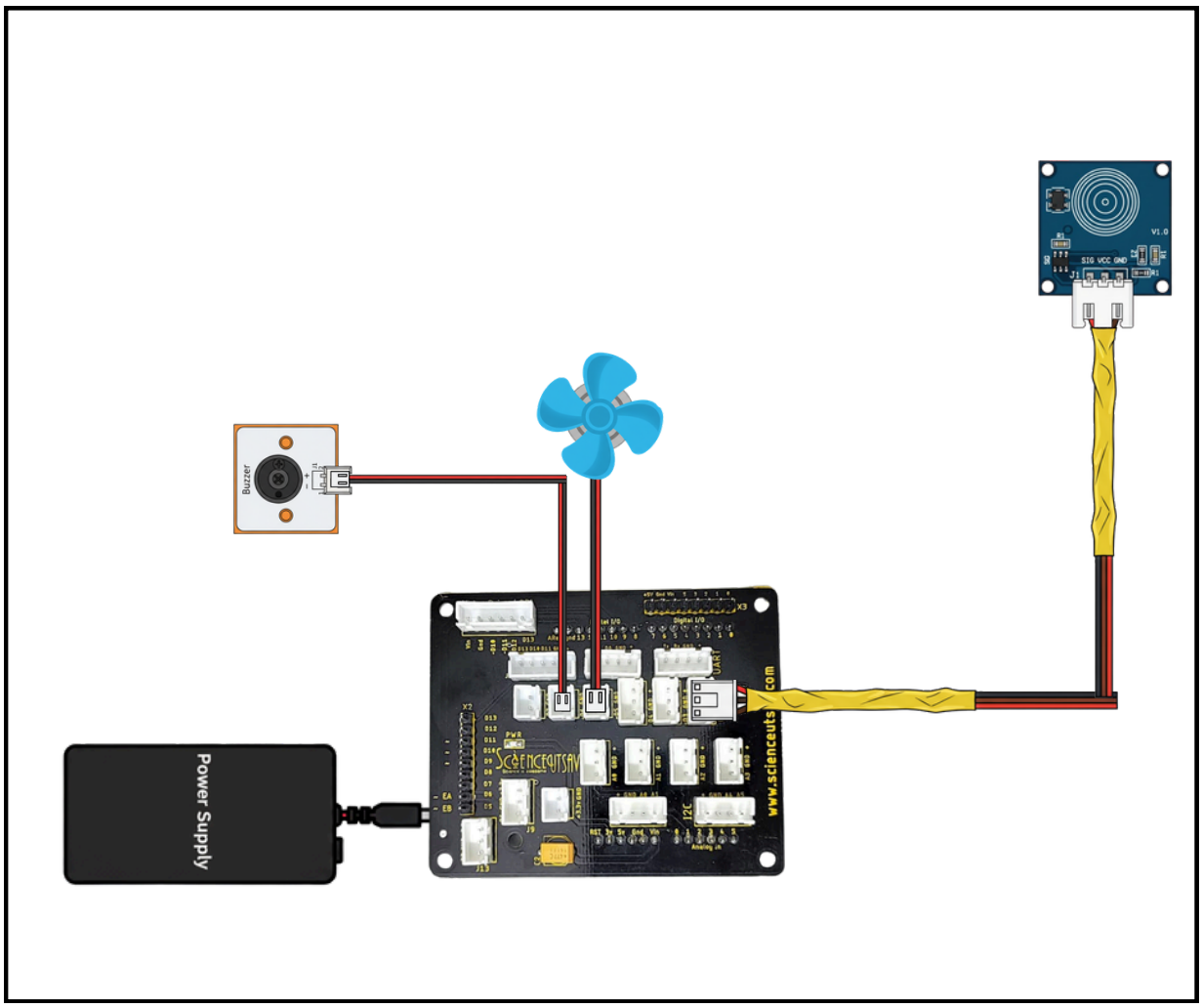
  if (speedLevel == 2) {
    analogWrite(dcMotor, 30);    // Medium speed
    Serial.println("Medium Speed");
  }

  if (speedLevel == 3) {
    analogWrite(dcMotor, 33);    // High speed
    Serial.println("High Speed");
  }

}
```

INTERMEDIATE PROJECT

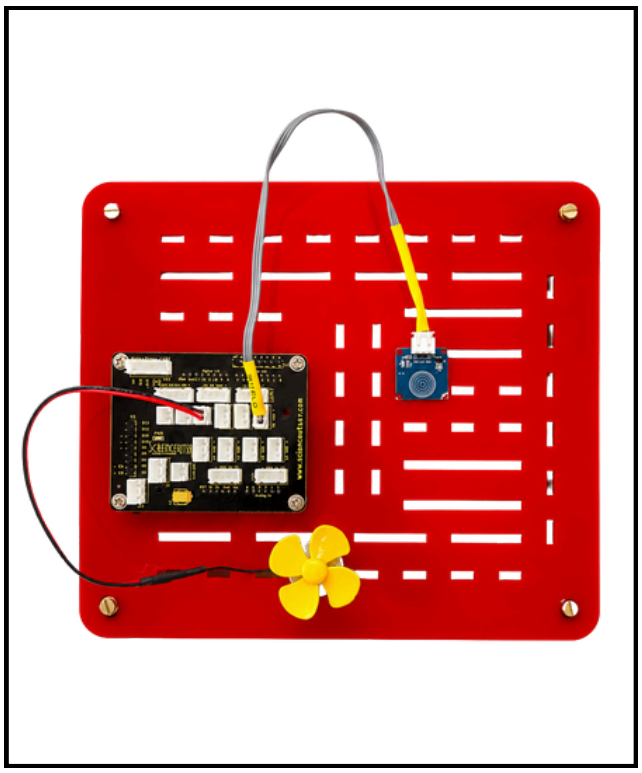
The circuit diagram of the project is given below



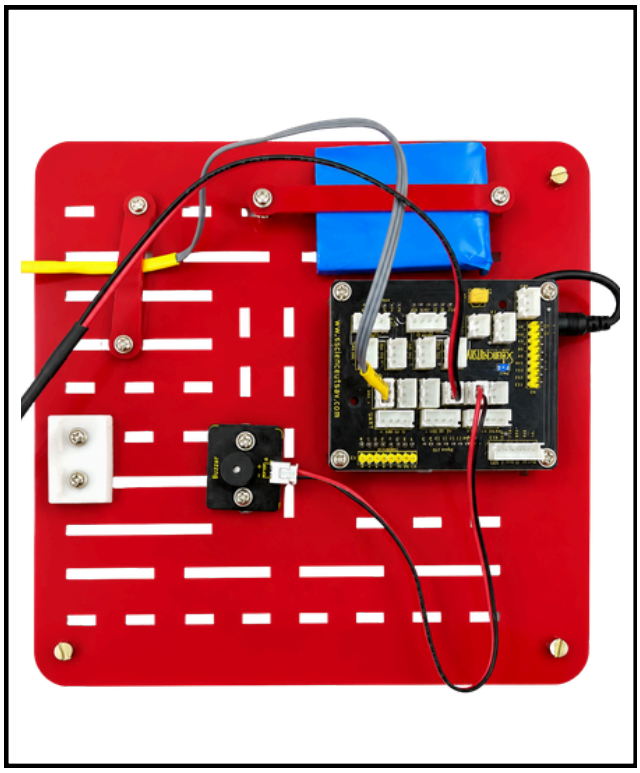
CONNECTIONS:

1. DC motor-D5
2. Buzzer-D6
3. Touch sensor-D2

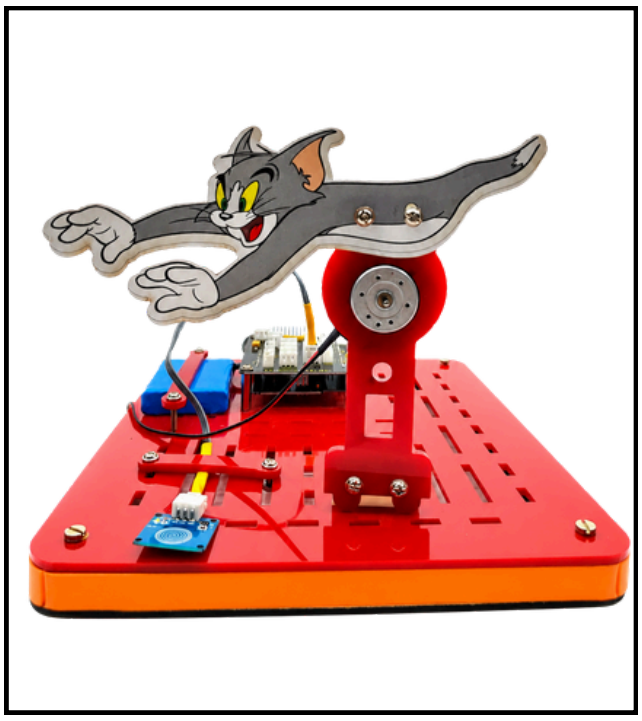
Step 1- Connect the touch sensor and DC motor to the board



Step 2- Connect the buzzer to D5 pin of shield.



Step 3- Attach the template to the stand as shown



Step 4- Fix the other template to the DC motor



Step 5- Upload the code and observe



Code

```
/*
ScienceUtsav Arduino Code
Touch Controlled Motor Speed with Continuous Cartoon Sound

Project: Tom & Jerry Tom Leg Movement Illusion

Touch Sensor = D2
DC Motor PWM Pin = D5
Buzzer = D6
*/

const int touchSensor = 2;
const int dcMotor = 5;
const int buzzer = 6;

int speedLevel = 0;
bool lastTouchState = LOW;

void setup() {
  pinMode(touchSensor, INPUT);
  pinMode(dcMotor, OUTPUT);
  pinMode(buzzer, OUTPUT);

  Serial.begin(9600);
}

void loop() {
  bool touchState = digitalRead(touchSensor);

  // Detect one touch at a time
  if (touchState == HIGH && lastTouchState == LOW) {
    speedLevel++;

    if (speedLevel > 4) {
      speedLevel = 0;
    }

    delay(300);
  }

  lastTouchState = touchState;
```

```
// Speed Level 0: Motor OFF
 if (speedLevel == 0) {
  analogWrite(dcMotor, 0);
  noTone(buzzer);
  Serial.println("Motor OFF");
 }

 // Speed Level 1: Slow motor + slow cartoon sound
  if (speedLevel == 1) {
    analogWrite(dcMotor, 28);

    tone(buzzer, 700);
    delay(120);
    tone(buzzer, 900);
    delay(120);

    Serial.println("Slow Speed");
  }

  // Speed Level 2: Medium motor + medium cartoon sound
  if (speedLevel == 2) {
    analogWrite(dcMotor, 30);

    tone(buzzer, 800);
    delay(90);
    tone(buzzer, 1100);
    delay(90);

    Serial.println("Medium Speed");
  }

  // Speed Level 3: Fast motor + fast cartoon sound
  if (speedLevel == 3) {
    analogWrite(dcMotor, 33);

    tone(buzzer, 900);
    delay(60);
    tone(buzzer, 1300);
    delay(60);

    Serial.println("High Speed");
  }
```

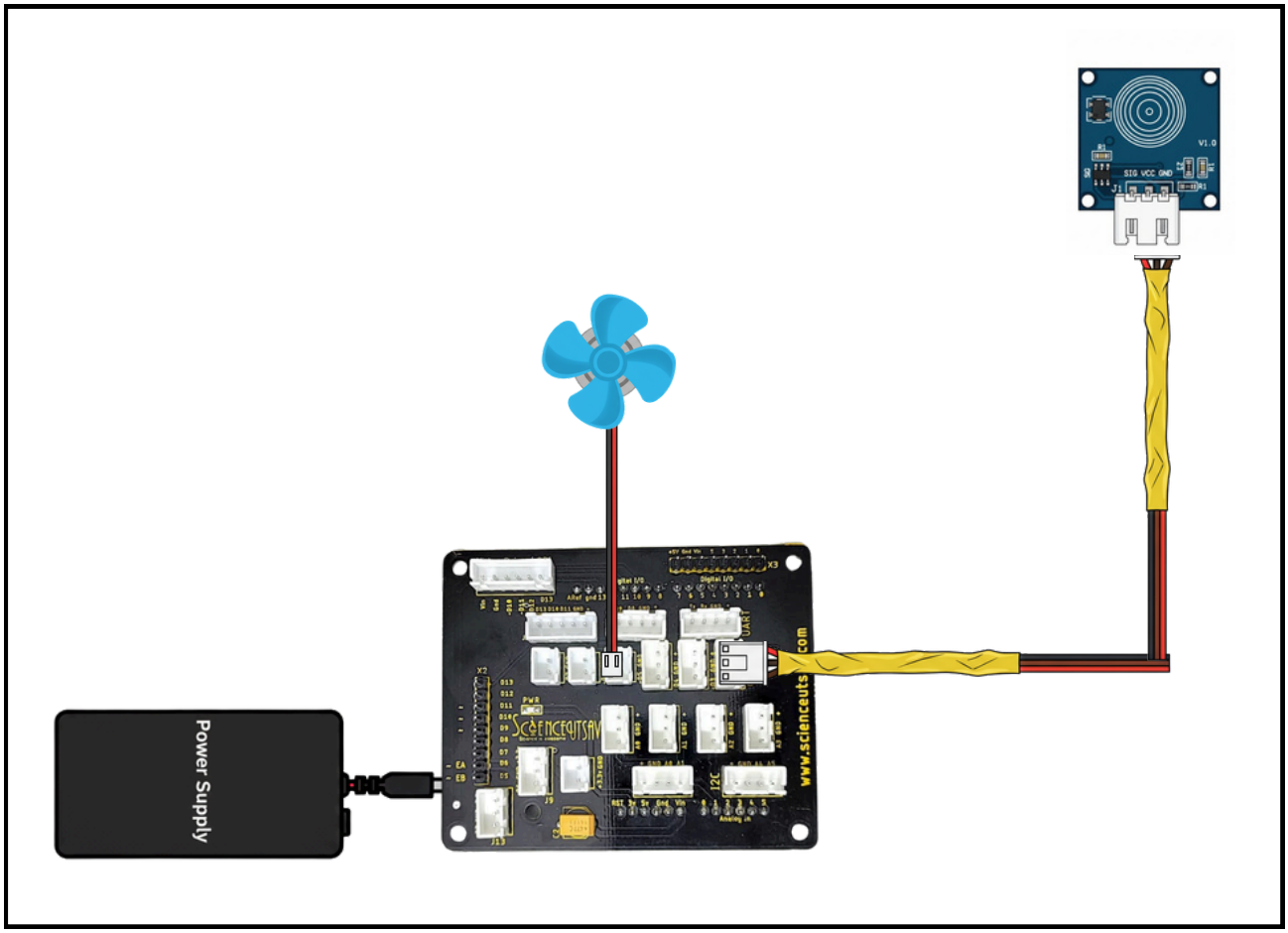
```
 // Speed Level 4: Full speed + very fast cartoon sound
 if (speedLevel == 4) {
  analogWrite(dcMotor, 35);

  tone(buzzer, 1000);
  delay(40);
  tone(buzzer, 1400);
  delay(40);
  tone(buzzer, 1800);
  delay(40);

  Serial.println("Full Speed");
 }
}
```

ADVANCED PROJECT

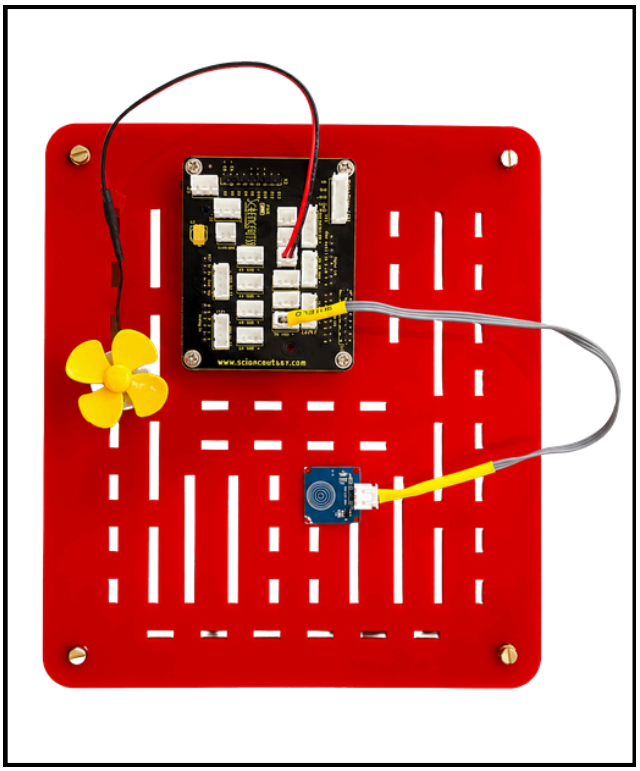
The circuit diagram of the project is given below



CONNECTIONS:

1. DC motor-D5
2. Touch sensor-D2

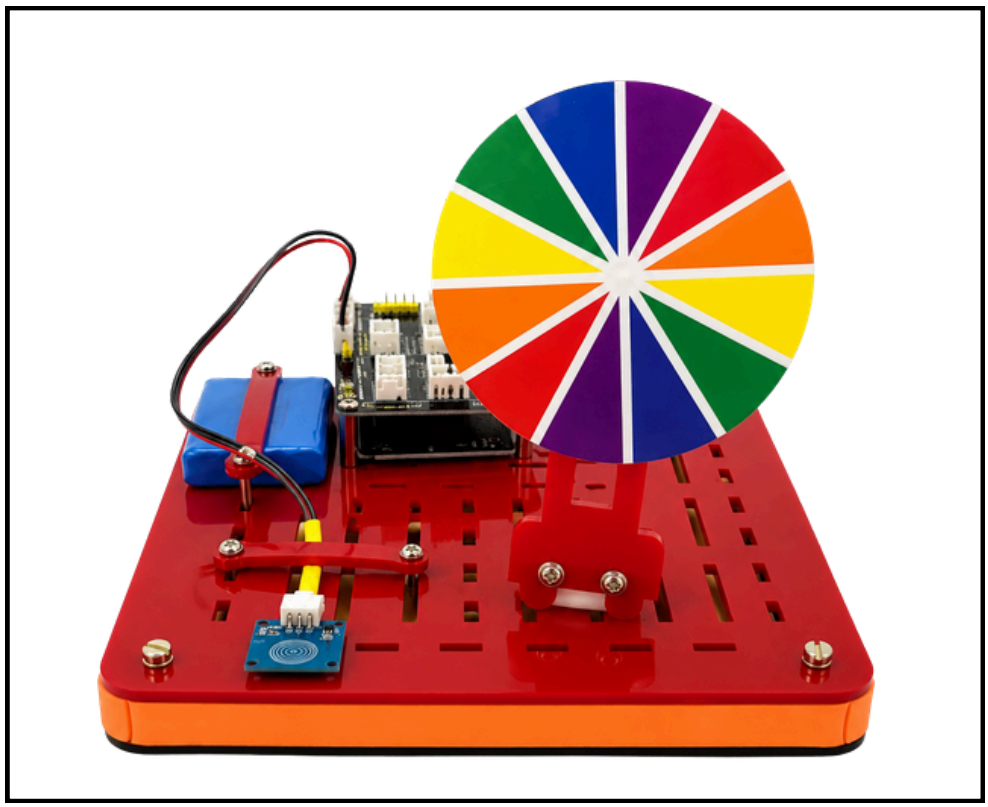
Step 1- Connect touch sensor and the DC motor as shown



Step 2- Attach the DC motor to the stand



Step 3- Attach the template to the motor



Code

```
const int touchSensor = 2;
const int dcMotor = 5;

int speedLevel = 0;
bool lastTouchState = LOW;

void setup() {
  pinMode(touchSensor, INPUT);
  pinMode(dcMotor, OUTPUT);

  Serial.begin(9600);
}

void loop() {
  bool touchState = digitalRead(touchSensor);

  // Detect one touch at a time
  if (touchState == HIGH && lastTouchState == LOW) {
    speedLevel++;

if (speedLevel > 3) {
   speedLevel = 0;
  }

  delay(300);
 }

 lastTouchState = touchState;
```

```
  if (speedLevel == 0) {
    analogWrite(dcMotor, 0);      // Motor OFF
    Serial.println("Motor OFF");
  }

  if (speedLevel == 1) {
    analogWrite(dcMotor, 27);     // Slow speed
    Serial.println("Slow Speed");
  }

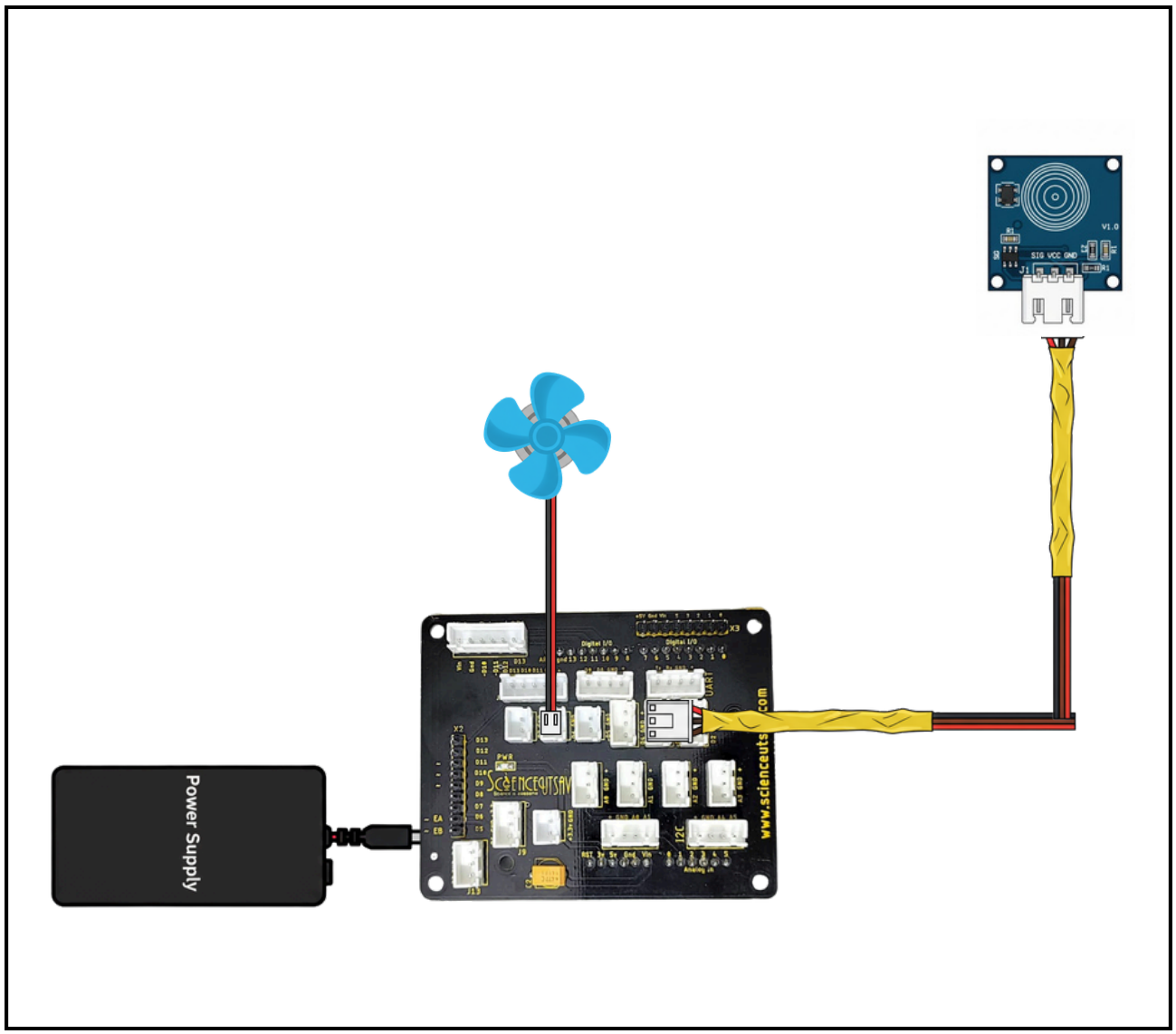
  if (speedLevel == 2) {
    analogWrite(dcMotor, 30);    // Medium speed
    Serial.println("Medium Speed");
  }

  if (speedLevel == 3) {
    analogWrite(dcMotor, 33);    // High speed
    Serial.println("High Speed");
  }

}
```

ADVANCED PROJECT

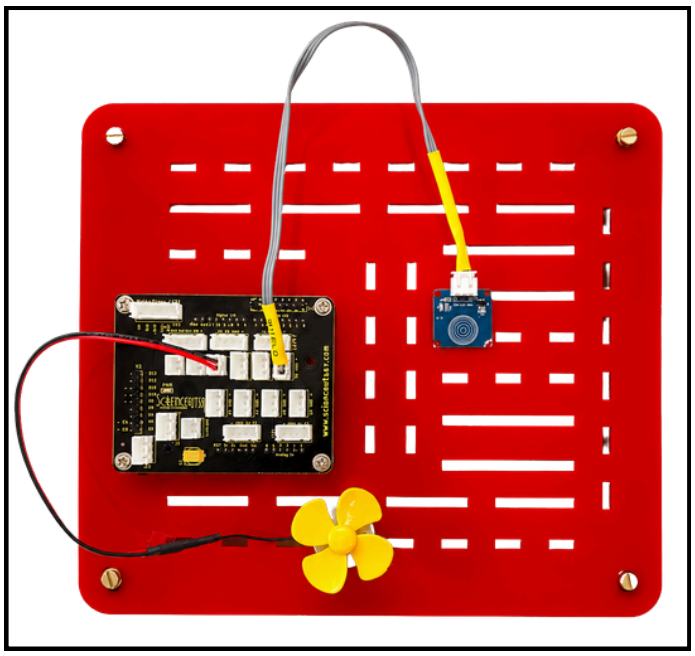
The circuit diagram of the project is given below



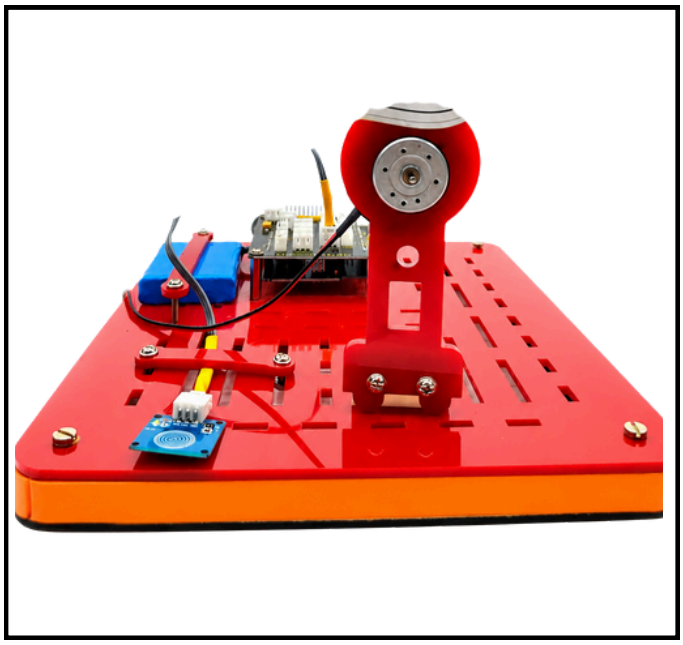
CONNECTIONS:

1. DC motor-D6
2. Touch sensor-D3

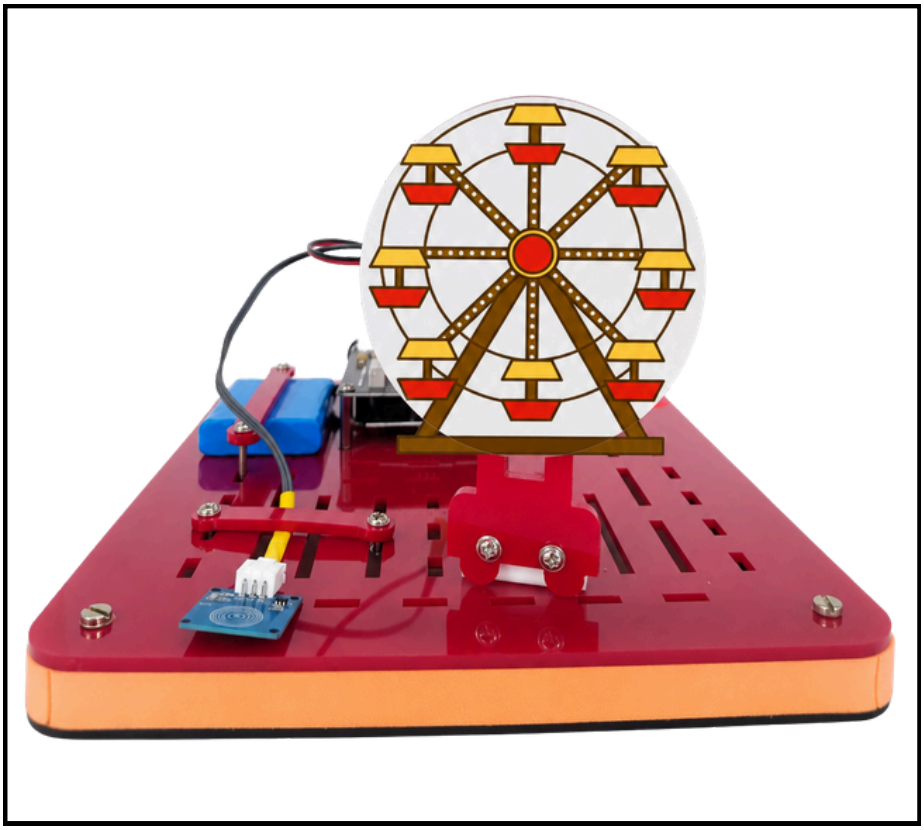
Step1- Connect touch sensor and the DC motor as shown



Step 2- Attach the DC motor to the stand



Step 3- Attach the template to the motor



Code

```
const int touchSensor = 2;
const int dcMotor = 5;

const int ldrSensor = 3;
const int led = 6;

// Change HIGH to LOW if LED works opposite
const int LIGHT_STATE = HIGH;

int speedLevel = 0;
bool lastTouchState = LOW;

void setup() {
  pinMode(touchSensor, INPUT);
  pinMode(dcMotor, OUTPUT);

  pinMode(ldrSensor, INPUT);
  pinMode(led, OUTPUT);

  Serial.begin(9600);
}

void loop() {
  bool touchState = digitalRead(touchSensor);

  // Detect one touch at a time
  if (touchState == HIGH && lastTouchState == LOW) {
    speedLevel++;

    if (speedLevel > 3) {
      speedLevel = 0;
    }
```

```
  delay(300);
  }

  lastTouchState = touchState;

   // Motor speed control
   if (speedLevel == 0) {
     analogWrite(dcMotor, 0);
     Serial.println("Motor OFF");
   }

   if (speedLevel == 1) {
     analogWrite(dcMotor, 27);
     Serial.println("Slow Speed");
   }

   if (speedLevel == 2) {
     analogWrite(dcMotor, 30);
     Serial.println("Medium Speed");
   }

   if (speedLevel == 3) {
     analogWrite(dcMotor, 33);
     Serial.println("High Speed");
   }

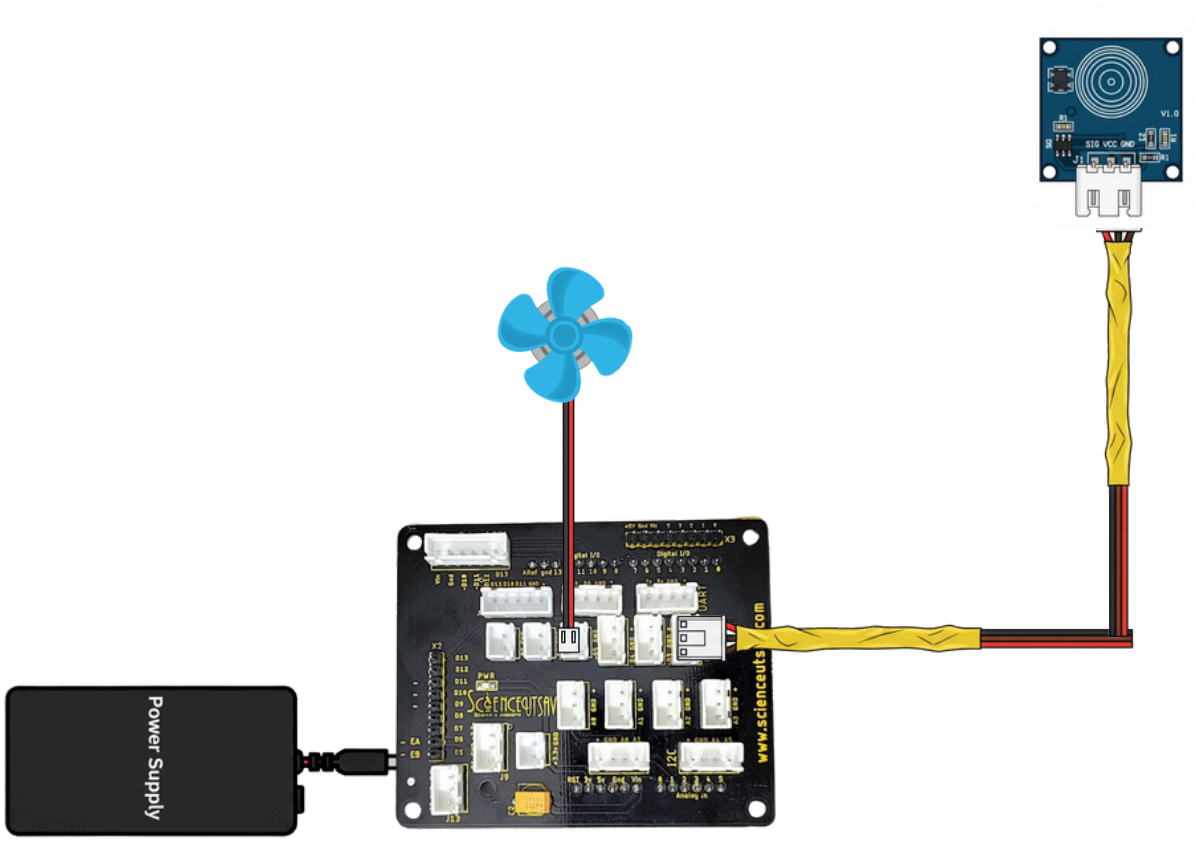
   // LDR light control LED
   bool lightDetected = digitalRead(ldrSensor) == LIGHT_STATE;

   if (lightDetected) {
     digitalWrite(led, HIGH);
   } else {
     digitalWrite(led, LOW);
   }

   delay(100);
 }
```

EXPERT PROJECT

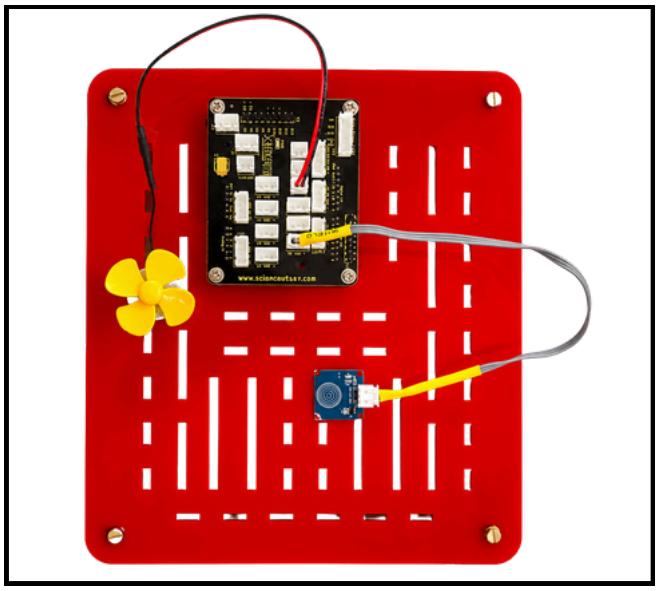
The circuit diagram of the project is given below



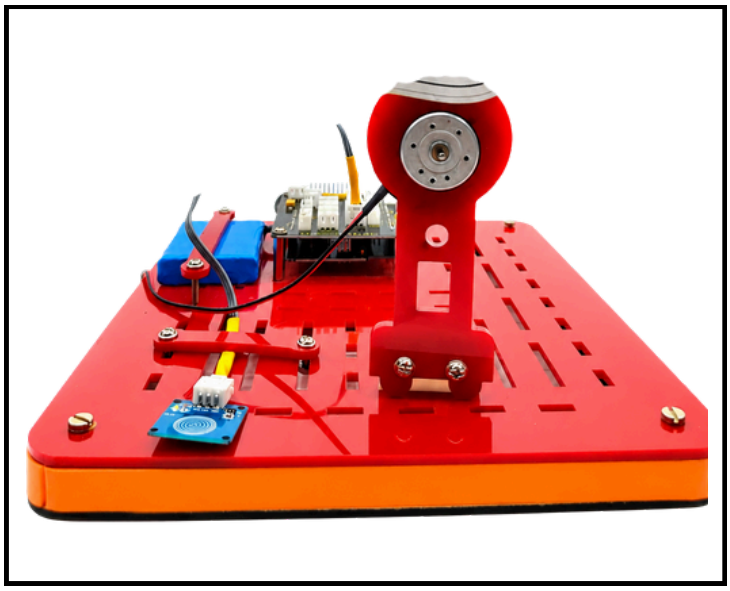
CONNECTIONS:

1. DC motor-D5
2. Touch sensor-D2

Step1- Connect touch sensor and the DC motor as shown



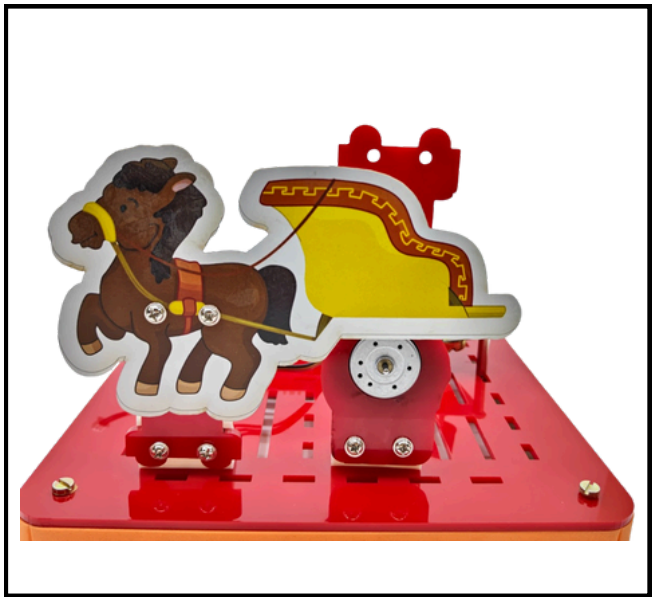
Step 2- Attach the DC motor to the stand



Step 3- Attach the template to the other stand



Step 4- Fix the stand as shown



Step 5- Attach the wheel to the motor



Code

```
const int touchSensor = 2;
const int dcMotor = 5;

int speedLevel = 0;
bool lastTouchState = LOW;

void setup() {
  pinMode(touchSensor, INPUT);
  pinMode(dcMotor, OUTPUT);

  Serial.begin(9600);
}

void loop() {
  bool touchState = digitalRead(touchSensor);

  // Detect one touch at a time
  if (touchState == HIGH && lastTouchState == LOW) {
    speedLevel++;

    if (speedLevel > 3) {
      speedLevel = 0;
    }

    delay(300);
  }
```

```
  lastTouchState = touchState;

  if (speedLevel == 0) {
    analogWrite(dcMotor, 0);      // Motor OFF
    Serial.println("Motor OFF");
  }

  if (speedLevel == 1) {
    analogWrite(dcMotor, 27);     // Slow speed
    Serial.println("Slow Speed");
  }

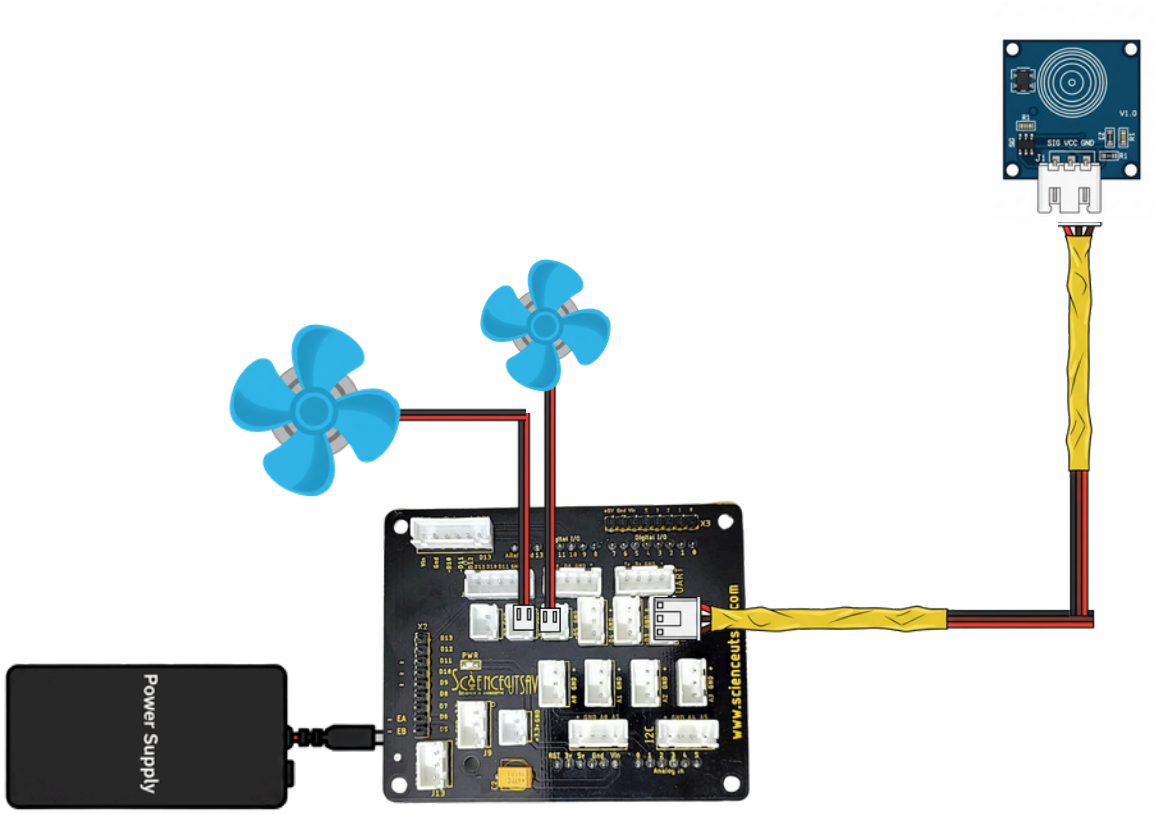
  if (speedLevel == 2) {
    analogWrite(dcMotor, 30);    // Medium speed
    Serial.println("Medium Speed");
  }

  if (speedLevel == 3) {
    analogWrite(dcMotor, 33);    // High speed
    Serial.println("High Speed");
  }

}
```

EXPERT PROJECT

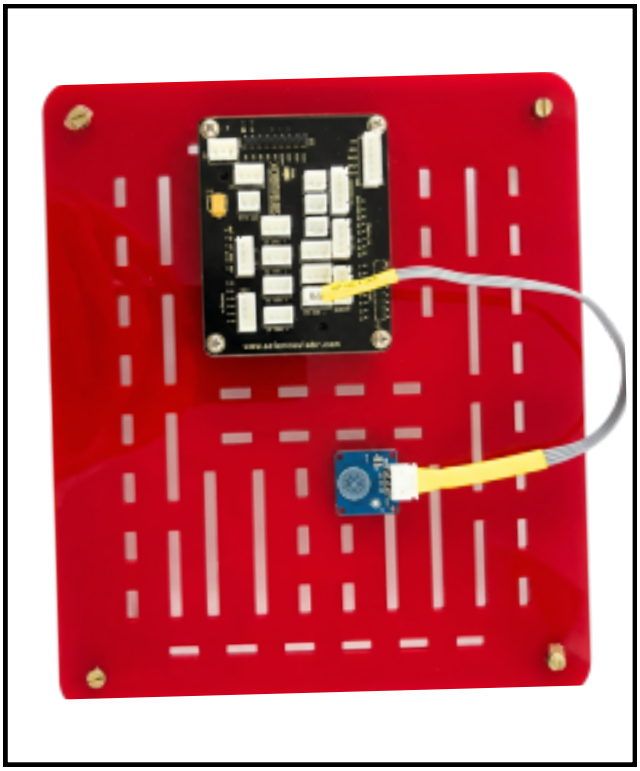
The circuit diagram of the project is given below



CONNECTIONS:

1. DC motor-D5, D6
2. Touch sensor- D2

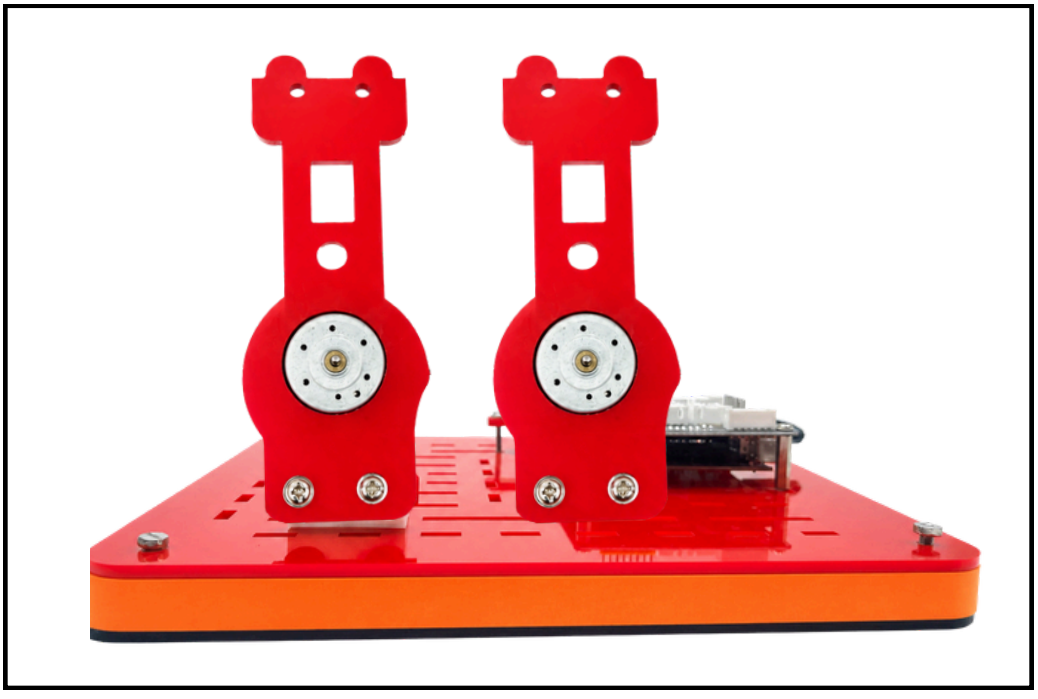
Step 1- Connect the touch sensor to the board



Step 2- Attach the DC motor to both the stands



Step 3- Fix the stands to the board as shown



Step 4- Attach the template and fix the wheels to the motors



Code

```
/*
ScienceUtsav Arduino Code - Session 6 to 10
Corrected for 3-pin DIGITAL LDR module.
LDR uses digitalRead(), not analogRead().
*/

const int touchSensor = 2;
const int dcMotor1 = 5;
const int dcMotor2 = 6;

void setup() {
  pinMode(touchSensor, INPUT);
  pinMode(dcMotor1, OUTPUT);
  pinMode(dcMotor2, OUTPUT);
}

void loop() {
  if (digitalRead(touchSensor) == HIGH) {
    digitalWrite(dcMotor1, HIGH);
    digitalWrite(dcMotor2, HIGH);
  } else {
    digitalWrite(dcMotor1, LOW);
    digitalWrite(dcMotor2, LOW);
  }
}
```